Second Law

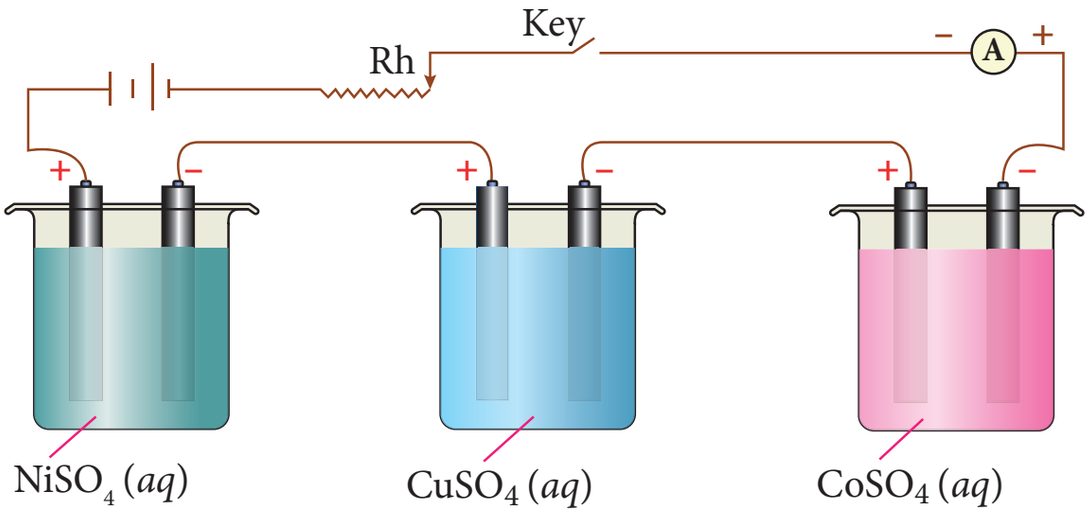


Figure 9.9 Electrolysis of various electrolytes using same quantity of charge

When the same quantity of charge is passed through the solutions of different electrolytes, the amount of substances liberated at the respective electrodes are directly proportional to their electrochemical equivalents.

Let us consider three electrolytic cells connected in series to the same DC electrical source as shown in the figure 9.9. Each cell is filled with a different electrolytes namely $NiSO_4$, $CuSO_4$ and $CoSO_4$, respectively.

When Q coulomb charge is passed through the electrolytic cells the masses of Nickel, copper and cobalt deposited at the respective electrodes be m_{Ni}, m_{Cu} and m_{Co} , respectively.

According to Faraday's second Law,

$$m_{Ni} \propto Z_{Ni},\ m_{Cu} \propto Z_{Cu} \text{ and } m_{Co} \propto Z_{Co}$$

(or)

$$\frac{m_{Ni}}{Z_{Ni}} = \frac{m_{Cu}}{Z_{Cu}} = \frac{m_{Co}}{Z_{Co}} \qquad \text{.....(9.36)}$$

Example

A solution of silver nitrate is electrolysed for 20 minutes with a current of 2 amperes. Calculate the mass of silver deposited at the cathode.

Electrochemical reaction at cathode is $Ag^+ + e^- \rightarrow Ag$ (reduction)

$m = ZIt$

$$m = \frac{108\ g\,mol^{-1}}{96500\ C\,mol^{-1}} \times 2400C$$

$m = 2.68$ g.

$$Z = \frac{\text{molarmass of Ag}}{(96500)} = \frac{108}{1 \times 96500}$$

$I = 2A$

$t = 20 \times 60S = 1200\ S$

$It = 2A \times 1200S = 2400C$

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